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Chapter 15: Dynamic Calibration of AI with MCAF—An Asymmetric Diagnosis of Projection-End Expansion and Defense-End Absence

Abstract

This chapter completes the dynamic calibration of the Macro-Cognitive Accounting Framework (MCAF) in the AI domain—advancing from static structural diagnosis to dynamic cumulative dynamics diagnosis. Chapters 13 and 14 respectively revealed the structural defects and ideological structures of AI systems; this chapter answers the dynamic questions that the preceding two chapters could not address: how much debt have these structural defects accumulated over time? Is the speed gap between the projection end and the defense end widening or narrowing? When will the settlement occur? This chapter takes as its starting point the controlled model-collapse research of Shumailov et al. (2024, *Nature*)—indiscriminate recursive use of generated data can produce distribution degeneration under specific protocols. When generated content returns to public corpora, its real-world scope still depends on the proportion of synthetic content, filtering, retention of real data, and successive training generations. AI's unlocked stock (C) therefore faces continuing accumulative pressure, and AI's debt is not merely a "stock problem" but can take the form of an "increment engine." On this premise, this chapter presents three structural diagnoses that can only be articulated within the MCAF framework: (1) AI's shadow debt (D_c^{shadow}) lacks the endogenous manifestation chain—"underlying asset default–margin call–rating downgrade"—that exists in the financial system; the concealment phase lacks the constraint of an institutionalized manifestation mechanism, making its duration highly uncertain. Even if the defense side accelerates,

AI still lacks an institutionalized endogenous channel that would make shadow debt manifest automatically. (2) The relation among the three clocks shows that, while current institutional and technological conditions persist, the defense side faces long-term lag pressure: the characteristic period of the techno-capital clock (τ_c) is measured in months, the governance clock (τ_g) responds on a yearly scale, and the debt clock (τ_d) has no stable current estimate. AI governance has not developed the stable emergency-compression capacity found in the financial system's crisis mode. (3) At the individual level, no institutional counterbalance against algorithmic sycophancy can be established—a cognitive system cannot simultaneously serve as its own independent overseer; this asymmetry in instantiation means that "user education" cannot substitute for institutional design. A state can use independent institutions to check power; an individual cannot instantiate an independent oversight body within their own cognitive system. The three diagnoses converge on a single conclusion: AI's shadow debt may be harder to track and harder to settle than financial shadow debt, its debt dynamics are structurally different from those of finance, and we have yet to develop auditing tools that match this difference. The contribution of this chapter is not to "solve" this structural predicament but to honestly annotate it—making the invisible accumulation of debt visible is the most fundamental work the MCAF framework can do in the AI domain.

Keywords

MCAF; cognitive debt; shadow debt; three clocks; unlocked stock; governance clock; techno-capital clock; institutional asymmetry; algorithmic sycophancy; local file audit

§1 Introduction

Chapter 13 completed, from the perspective of static structure, the "4+1" diagnosis of AI by the fiber framework—the four-layer progressive structure: how long the mapping chain is (L0), where optimization is unconstrained (L1), whether a certain appearance conceals an uncertain substance (L2), and what happens when multiple mapping systems reference one another (L3); as well as the narrative load product (L4) as an emergent layer. Chapter 14 further unfolded the ideological structure of the discursive legitimacy layer of L4—Western-centrism, individualism, and the narrative of silicon-based civilization. But static diagnosis cannot answer three dynamic questions: how much debt have these structural defects accumulated over time? Is the speed gap between the projection end and the defense end narrowing or widening? When will the settlement occur?

This chapter uses the MCAF framework to answer these three questions. §2 presents the background facts of the AI industry and takes three fundamental differences in debt structure between AI and finance as the empirical premises for this chapter's diagnosis—among them, the proposition that indiscriminate recursive use of generated data can produce distribution degeneration is anchored by the controlled model-collapse research of Shumailov et al. (2024). When the chapter further discusses generated content returning to public corpora, the proportion of synthetic content, filtering procedures, retention of real data, and successive training generations must remain conditions for investigation; the controlled result is not treated as proof that irreversible degeneration has already occurred throughout the training ecology. On this premise, §§3–5 sequentially unfold three structural diagnoses that can only be articulated within the MCAF framework: why shadow debt lacks an institutionalized endogenous manifestation chain (§3), why the relation among the three clocks places the defense side under long-term lag pressure under current conditions (§4), and why, at the individual level, no institutional

counterbalance against algorithmic sycophancy can be established (§5). §6 provides the falsification conditions for these framework inferences, and §7 concludes.

§2 Background Facts

Before unfolding the diagnosis, the background facts of the AI industry are presented. The presentation of these facts does not use MCAF framework variables.

Model iteration is accelerating. The Stanford HAI 2026 AI Index shows that in 2025, industry produced over 90% of notable frontier models. OpenAI, Anthropic, Google, DeepSeek, Qwen—each series iterates within months. The same report shows that generative AI reached a 53% population-level adoption rate within three years, a diffusion speed exceeding the historical pace of the PC and the internet; the organizational adoption rate rose to 88% (from the McKinsey 2025 Global AI Survey cited in the report, where "adoption" refers to an organization using AI in at least one business function). When alternative channels in the information environment are systematically compressed, "not using AI" shifts from an optional exit to a counter-default behavior requiring active resistance.

Regulatory responses are catching up, but the speed gap remains significant. The EU AI Act took roughly three years from its 2021 proposal to its 2024 enactment. China's interim measures for generative AI management were issued in 2023, taking roughly six months (Cyberspace Administration of China et al., 2023). The United States lacks comprehensive AI legislation at the federal level, relying primarily on executive orders and enforcement by existing agencies. The governance clocks across jurisdictions range from six months to three years or longer.

Accidents are increasing. The Stanford HAI 2026 AI Index recorded AI accidents rising from 233 in 2024 to 362 in 2025, an annual growth rate of approximately 55%. But these are only the recorded accidents—the long-term decline of meta-audit capacity (A), the systematic replacement of independent calibration behavior by frictionless interfaces, and the erosion of institutional trust—these chronic, diffuse debt accumulations are not in any accident statistics. AI has yet to experience a "Lehman moment."

There are three fundamental differences in debt structure between AI and finance. First, model outputs can flow back into subsequent training data, producing self-reproduction of training data. Shumailov et al. (2024) show that, under specific protocols of indiscriminate recursive use of generated data, output diversity and accuracy decline across iterations and information in the distribution tail can be irreversibly lost. The scope of this mechanism in the real training ecology still depends on the proportion of synthetic content, filtering procedures, and retention of real data. Financial credit stocks contract after a bubble bursts; recursive training without the calibration described above can continue to accumulate degenerative pressure. This comparison does not mean that all AI data contamination is irreversible or incapable of mitigation. Second, AI's narrative load is a second derivative—not specific false assertions, but the meta-rules defining "what counts as a reasonable narrative." The narrative of silicon-based civilization elevates AI from "tool" to "civilizational subject," eliminating the legitimacy of human auditing at the discursive level. Financial narratives are primarily first-order—manifested as concrete, auditable narratives ("house prices will only rise"). First-order narratives can be refuted by facts; second-order narratives define "what is a fact" (Chapter 14, §§3–4 have argued this in detail). Third, finance's legitimacy base, having evolved over centuries, is relatively stable—deposit insurance, central bank lender of last resort, and bankruptcy law constitute an institutionalized compensation loop. AI currently has no institution-

alized compensation loop—there exist scattered product liability remedies and case-by-case litigation, but no scaled, automatically triggered compensation/liquidation mechanism corresponding to deposit insurance, lender of last resort, and bankruptcy law. When the actions of an AI Agent cause loss, the loss distribution pathway either relies on case-by-case judicial action or directly impacts the end user. This is not a problem of "the product not being good enough"; it is a problem of "there are no rules for handling things when they go wrong."

The following three diagnoses are structural insights that can only be articulated under the MCAF framework. Each satisfies the test criterion: if the MCAF variables were removed, could this passage still reach the same judgment? If not, then the framework is doing genuine work.

§3 Diagnosis One: Shadow Debt—Why the Concealment Phase Lacks Institutional Constraint

Shadow debt (D_c^{shadow}) in the MCAF framework is defined as the gap between true cognitive debt and observable cognitive debt. In the financial system, shadow debt is ultimately forced into visibility through underlying-asset defaults, rating downgrades, and margin calls. No comparable financial-style compulsory manifestation mechanism has yet formed in AI systems. The manifestation function $M(\tau)$ named in Chapter 3, §4.2 marks the scenario-specific transformation through which shadow debt becomes observable; what the AI domain lacks is not manifestation itself, but an institutionalized endogenous chain that makes it occur automatically—the three conditions unfolded below are the components of this chain.

This is not an empirical observation—"AI has not yet experienced a large-scale accident triggering settlement"—but a structural judgment. The reason financial shadow debt can be forcibly manifested depends on three conditions holding simultaneously. First, underlying assets exist—the underlying assets of mortgage-backed securities (MBS) are mortgages, and the underlying assets of mortgages are the borrowers' repayment capacity. When house prices fall, default rates rise; this is a physical causal chain that cannot be concealed by narrative. Second, a margin system exists—when asset prices fall below a threshold, margin calls are automatically issued, independent of anyone's judgment. Third, rating agencies exist—they, though riddled with conflicts of interest, at least formally provide risk assessments independent of the issuer. When ratings are downgraded, institutional investment portfolios are forcibly adjusted, and a portion of shadow debt is forcibly converted into observable losses.

AI currently possesses none of these three conditions. There are no underlying assets—a rise in a model's "hallucination rate" does not automatically trigger anyone's margin call. The "information cocoons" of recommendation algorithms do not lead to any institution's rating downgrade. Attention fragmentation and judgment degradation are real losses, but they do not automatically generate price signals, margin mechanisms, or external audits with compulsory force. They are diffuse, chronic, and difficult to attribute, and therefore lack a moment of settlement automatically recognized by an institution.

This gap leaves traces on externally observable indicators: after physical co-presence spaces (office buildings, street-level shops) have continuously given way to online platforms, the freed attention has not automatically been converted into independent judgment capacity but has been recaptured, in more fragmented forms, within the multiplicative channel of the projection end ($\Lambda \cdot \phi$)[1]—the quantification of this form still lacks direct indi-

[1]The basic premise is the constraint established in Chapter 7—the information throughput consciously processed by an individual is on the order of 10 bits/s, a hard cognitive scarce resource; when the temporal distribution of this finite budget undergoes a structural migration, the physical infrastructure that receives it—office buildings, street-level shops—records this migration in the form of vacancy rates. Moody's Analytics (2025) data shows that the U.S. office vacancy rate reached a historical peak of 20.6% in the second quarter of

cators; this section merely takes vacancy rates as a multi-cause, single-effect external reference indicator and does not use them to establish a single causal chain.

This reveals the core characteristic of AI shadow debt: the concealment phase lacks the constraint of an institutionalized manifestation mechanism, so its duration is highly uncertain. Financial shadow debt can accumulate in the dark for years—the subprime accumulation from 2003 to 2007—but will ultimately be exposed by the default rate of underlying assets. AI shadow debt has no comparable institutionalized endogenous outlet. It can only be exposed under two conditions: either users abandon AI services en masse, voting with their behavior—but this is not an economic mechanism; it is a behavioral change, and its time scale is unpredictable; or a sufficiently large accident forcibly triggers regulatory intervention—and currently, no institutionalized triggering mechanism exists to make such intervention inevitable. The former is logically possible but temporally unpredictable; the latter—an institutionalized triggering mechanism—simply does not exist in the current AI governance architecture.

In the language of MCAF variables, this gap in the manifestation chain is equivalent to the non-closure of the legitimacy base ($\|L\|$): finance's compensation loop—deposit insurance, lender of last resort, bankruptcy law—is the institutional carrier of $\|L\|$ in the financial domain, providing legitimacy support for loss distribution and settlement after shadow debt is manifested; AI currently has no such institutional carrier, so "the absence of the shadow debt manifestation chain" and "the non-institutionalization of $\|L\|$" are two projections, onto different MCAF variables, of the same structural gap—the former manifests at the level of the dynamics of D_c^{shadow}; the latter manifests at the level of the institutional base of $\|L\|$. This duality is the mirror gap, in the AI domain, of Chapter 18 (compensation systems and $\|L\|$): finance's $\|L\|$, having evolved over centuries, is relatively stable; AI's $\|L\|$ has not yet begun to accumulate.

A Reverse Reference from Environmental History. The concealment phase of finance is measured in years (the subprime bubble 2003–2007), but environmental history provides a precedent orders of magnitude more extreme, and its settlement path precisely exposes the structural predicament of AI shadow debt. The record of London coal-smoke pollution can be traced back at least to the 13th century—Corton (2015), citing Brimblecombe (1987), confirms that London had street-name evidence of coal imports as early as 1228; 17th-century legislative attempts (a ban passed by the House of Lords in 1623) failed for lack of enforcement. For more than three centuries thereafter, coal smoke diffused through the city's daily life—London's elevated death rate had long been recorded in the Bills of Mortality, and Graunt's 1662 observations had already linked sea-coal smoke to mortality (Brimblecombe, 1987); the fog of December 1873 caused approximately 700 additional deaths within a single week—yet institutional response remained absent. The concealment phase lasted approximately seven hundred years.

The Great Smog of December 5–9, 1952, ended this concealment—the five-day fog episode, superimposed with the delayed deaths of the following two months, caused approximately 12,000 excess deaths (Bell & Davis, 2001; Bell et al., 2004), far exceeding the 3,000–4,000 generally reported at the time; the mortality data was obscured for nearly fifty years by the alternative attribution to an "influenza epidemic," until the recalculation by

2025, setting a new record high once again. The drivers of vacancy rates are more than one—interest rates, remote work policies, and changes in consumption and office patterns can all be listed as candidate explanations; this section does not assert a single cause: among these candidates, only algorithms and interest rates act along the same channel on the **current and future distribution** of attention, while remote work policies, etc., are not on the same plane as the MCAF diagnostic chain. Vacancy rates are taken as a multi-cause, single-effect reference indicator rather than a deterministic inference: after physical co-presence spaces have continuously given way to online platforms, the freed attention has not automatically been converted into independent judgment capacity.

Bell & Davis (2001) restored the true scale. The then Housing Minister Macmillan, in public statements, insisted that "I am not satisfied that further general legislation is needed at present," delaying a legislative response on the grounds of "the enormous number of broad economic considerations" (GLA, 2002; Davis, 2002; Corton, 2015). Nevertheless, the 1952 event ultimately forced the passage of the Clean Air Act of 1956.

The reason the London fog ultimately had a "settlement moment" is that its cost could be counted in bodies—12,000 excess deaths constituted undeniable physical evidence, even if that evidence was seven hundred years late. In the language of MCAF, the manifestation function $M(\tau)$ of the London fog ultimately underwent a forced jump. AI's cognitive harms—attention fragmentation, judgment degradation, the diffuse decline of independent calibration behavior—will not, within any five-day window, stack up into a visualizable mortality curve. The diagnosis of this section is therefore sharper than the analogy of the London fog suggests: the AI domain still lacks a physical trigger comparable to body counts that automatically constrains institutional response[2].

§4 Diagnosis Two: The Ordering of the Three Clocks—Why the Defense Side Lags under Current Structural Conditions

The three clocks of the MCAF framework provide a precise language for this diagnosis. The techno-capital clock (τ_c) marks the characteristic period over which the projection-side product changes significantly; competitive selective pressure favors faster-growing participants and may compress that period to a scale of months. The governance clock (τ_g) is constrained by the four sub-processes of identification, verification, formulation, and implementation and is measured in years. The debt clock (τ_d) marks the characteristic delay from the sedimentation of cognitive debt to its first manifestation.

What can currently be compared directly is that the techno-capital clock (τ_c, 1–3 months) is shorter than the governance clock (τ_g, 6 months–3 years). No stable estimate is available for the debt clock (τ_d); during concealment it is generally longer than the other two, but this does not justify a fixed numerical ordering. This is the AI version of the Collingridge Dilemma—when intervention is easy, the problem is not yet visible; when the problem is visible, intervention is already difficult.

An Honest Annotation of the Scaling Property: The above time-scale assessment (techno-capital clock 1–3 months, governance clock 6 months–3 years, and no stable current estimate for the debt clock) reflects **orders of magnitude from empirical observation**, not results calculated from the optimization dynamics equations of the MCAF framework—the MCAF framework does not internally contain a mechanism for predicting the specific numerical values of these three clocks; it proposes only a **structural tendency in their ordering** while current institutional and technological conditions persist. The specific scales are empirically given by industry data (the model release cycles of major AI labs), observations of legislative cycles (the passage cycles of AI legislation in major jurisdictions), and model collapse research. Therefore, the strength of the diagnosis of "long-term lag

[2]"Algorithmic pollution" as an academic concept already has independent theoretical work: Marjanovic, Cecez-Kecmanovic & Vidgen (2021), in the *Journal of Information Technology*, defined the systemic harm caused by automated algorithmic decision-making (AADM) to vulnerable groups as a new type of social pollution, with the subtitle "Making the Invisible Visible" sharing the same form as the programmatic goal of this chapter. But the "invisibility" to which the two refer points to different levels: the invisibility discussed by Marjanovic is institutional—black-boxed by organizations, legitimized by the discourse of "objectivity"—and is in principle discoverable by audit and attributable to specific individuals and specific algorithms; the invisibility discussed in this section is epistemological—an institutionalized endogenous structure that would automatically manifest D_c^{shadow} does not exist. The two works corroborate each other in the task of "making the invisible visible," but the diagnostic level of the MCAF framework (the dynamic gap of shadow debt) cannot be reduced to the first-order AADM harm attribution problem within the Marjanovic framework.

under current structural conditions" depends on two independent statements: the **ordering tendency** (given jointly by MCAF's structural constraints and the Collingridge Dilemma, an intra-framework inference) + the **empirical scales** (given by industry data and observations of legislative cycles up to 2026, external empirical premises). The two can be falsified separately: the falsification condition for the ordering relationship is given in §6 (if τ_g/τ_c continuously shrinks to less than 1); the correction of the empirical scales occurs continuously with new industry data, and the second edition of this book is expected to provide a recalibration up to 2028–2029. The layered annotation of intra-framework inferences and external empirical premises is to ensure that the diagnosis of the ordering relationship still holds when the specific clock values are dynamically updated with the industry.

But the incremental insight of the MCAF framework goes further than this. The governance clock of the financial system has a bimodal distribution in crises—slow in normal times, extremely fast in crises. Within weeks of the Lehman bankruptcy in September 2008, the Federal Reserve and the Treasury launched the Troubled Asset Relief Program (TARP), the American International Group (AIG) bailout, and the Commercial Paper Funding Facility, with decision-execution cycles measured in days. This is not the normal mode of the governance clock but an emergency intervention capability unique to the financial system—central banks can inject liquidity into the market within hours.

Current AI governance has not developed an emergency mode institutionalized to the same degree as that of finance, and no single rule change can instantly stop all data centers from operating. Under the current institutional architecture, the AI governance clock operates mainly on a monthly-to-yearly scale and has not demonstrated a stable emergency-compression capacity comparable to financial crisis response. This means: even if the defense end begins to accelerate, the already accumulated shadow debt still has no institutionalized manifestation channel. Finance's emergency mode can compress the governance clock from years to days—this emergency mode is not the product of design but the product of crises: from the Federal Reserve Act of 1913 to the Federal Deposit Insurance Corporation (FDIC) in 1933 to TARP in 2008, each was established only on the ruins of the previous crisis. Under the existing institutional architecture, the AI governance clock is difficult to compress to the same scale and remains primarily in a position of catching up.

More critically, the techno-capital clock in the financial system is constrained by the expansion speed of physical infrastructure—fiber optic laying, exchange expansion, data center construction—which requires years to a decade. The **short-term** expansion channel of the techno-capital clock (τ_c) in the AI domain is primarily on the software side—the visible timelines of model iteration, weight updates, and application programming interface (API) deployment are far shorter than the physical timelines of financial infrastructure construction. AI is not without a material substrate (the positive feedback constraints of computing power, electricity, supply chains, and planetary boundaries are detailed in Chapter 32), but at the visible timeline of the "capital–iteration–deployment" layer, the iteration rhythm on the software side is not directly constrained by the pace of physical engineering. The incompressible lower bound of the governance clock and this short-term degree of freedom of the techno-capital clock together constitute a structural pressure unique to AI—a pressure that does not exist in the financial domain.

§5 Diagnosis Three: Institutional Asymmetry—Why Individuals Cannot Resist Algorithmic Sycophancy

The preceding two sections completed the dynamic diagnosis of the MCAF framework at the macro level. Now, a deeper question is pursued: why can the expansion of the projection end advance almost unresisted at the individual level?

This is not a technical problem but an institutional one. At the individual level, there is a lack of institutional counterbalance against sycophancy.

A state can establish independent oversight bodies. The ancient Chinese oversight system included the censorate system (yushi), responsible for inspection and impeachment, and the remonstrance system (jianguan), responsible for remonstrance and the review and rejection of imperial edicts; the two complemented each other to form a composite oversight system (Zhang, 2017). The Qin and Han Dynasties already had a structure in which censors and remonstrators coexisted, and the Sui and Tang Dynasties brought it to full maturity. In the Roman Republic, the cursus honorum (the sequence of offices) prescribed the hierarchical sequence of offices—quaestor, aedile, praetor, consul—that administrative officials had to pass through, each office held for one year. Among the consuls (two in number) and among the tribunes, mutual veto power (intercessio) could be exercised. The core design principle of these systems is the same: **establish an independent oversight body whose interests do not overlap with those of the overseen object.** A state can do this because a state is an organization—it can establish a department whose budget, personnel, and evaluation are all independent of the department being overseen.

An individual cannot instantiate this "independent oversight body" within their own cognitive system—viewed from the fiber framework, the self-checking operation of any cognitive system cannot be independent of that system's mapping function (i.e., it cannot calibrate itself from the outside). This is structurally isomorphic with the organizational independence of a state oversight body relative to the overseen object: the loss of state oversight independence (the merger of censorate and remonstrance) deprived the self-supervision of power of its last institutional confrontational structure; "internal oversight" at the individual level is impossible to instantiate in the first place, not because "people are not smart enough," but because of an asymmetry in instantiation—a cognitive system can perform limited metacognitive self-checks, but it cannot instantiate, within itself, an oversight body whose budget/personnel/evaluation are independent of the overseen object in the sense of organizational independence. To simultaneously be the executor and an overseer independent of the executor in the organizational sense—this structure cannot be instantiated at the level of individual cognition.

Research on the Persuasion Knowledge Model has found that an individual's resistance to persuasion relies on persuasion knowledge accumulated over time—identifying when, how, and why a persuader attempts to influence them, and judging the manipulativeness of their tactics (Friestad & Wright, 1994). On this basis, this chapter adds a second condition: perceiving one's own vulnerability to that manipulation. But both conditions are systematically weakened in the algorithmic environment. Algorithmic manipulation has no single subjective "intent," but this does not mean it is unstructured: at the model level, it is the asymptotic dynamics of reward function maximization (Chapter 10 termed this "xiangyuan," the mathematical inevitability of an optimization process satisfying specific conditions); at the product level, it is the commercial choice of companion AI—the interface preference for low friction, dependence, and sustained dwell is written into the reward function itself (Chapter 10 termed this "sycophancy"; see the comparison table in Chapter 10, §5 for details; Weizenbaum,

1976; Muldoon & Parke, 2025). Users also do not perceive "vulnerability"—the helpful, honest, and harmless (HHH) product eliminates discomfort, and discomfort is the signal of vulnerability. The review published by Zheng and Meister (2025) in *Neuron* bounds the information throughput of human conscious behavior at approximately 10 bits/s. When AI's output speed is calculated in thousands of tokens per second, what the individual faces is a structural asymmetry: counterbalance at the state level is possible because the state can establish an institutional oversight body independent of the overseen object; counterbalance at the individual level is impossible because the organizational conditions required to establish such a body do not hold at the individual level.

This means that "user education" or "improving digital literacy" cannot substitute for institutional design. It is not that these efforts are valueless, but that they cannot solve the structural problem. Even if a user fully knows that the algorithm is manipulating their attention, they do not have an independent internal oversight body to counterbalance this manipulation at every decision moment.

An Engineering Precedent from the Game Industry UX. The judgment that "the HHH product eliminates discomfort" already has a long history of systematic engineering practice in the field of game user experience (UX). Celia Hodent, in *The Gamer's Brain: How Neuroscience and UX Can Impact Video Game Design*, published by CRC Press in 2017, systematically translated over a hundred experimental conclusions of cognitive neuroscience and UX research into an actionable diagnostic checklist for game development teams—covering five modules: perception, attention, memory, motivation, and emotion. Hodent opens by analogizing game design to magic: magicians—and game designers—succeed precisely because they "learn to take advantage of brain loopholes" and master specific practices such as "misdirection" (diverting the audience's attention to deceive perception), ultimately causing players to enter a state of "suspend their disbelief" (Hodent, 2017). Most directly relevant to this chapter's thesis is her treatment of cognitive biases: cognitive biases, like visual illusions—"no matter how aware of these biases you are, you will fall prey to them" (Hodent, 2017). This is completely isomorphic with the argument of this section: even if users perceive the existence of manipulation, they cannot escape being manipulated by virtue of that awareness—perceiving vulnerability does not equal being able to resist it.

The diagnostic value of this industry precedent is twofold. First, according to the positioning of this book, Hodent translated cognitive neuroscience knowledge into actionable processes for development teams, providing the game industry with a complete set of design tools for systematically eliminating players' feelings of friction and frustration. Viewed from the fiber framework, this is precisely the complete engineering of the "HHH product" in the game industry: discomfort is systematically measured, located, and eliminated, and this set of methods is written into a textbook as a mainstream tutorial. Second, Hodent deals with entertainment products; when the same UX methodology is carried into AI products centered on large language models (LLMs), with subscriptions, default pricing, and dark patterns as their business model, and high-emotion real-time interaction as their selling point, what is eliminated is no longer the game player's frustration at getting stuck but the user's surface-level discomfort with sycophancy, bias, and manipulation. Hodent's book was written from a stance of responsibility toward game developers and was not prepared to answer, on the ethical dimension, the question of "what happens when this set of methods is migrated to a platform where the cost of user addiction is higher." The judgment of this chapter is: this second-layer migration is already occurring, and it will not be resolved by individual self-resistance—in a situation where a CRC Press-level tutorial had already confirmed as early as 2017 that "even if aware of the biases, one cannot avoid falling prey to them," adding the capacity for cognitive

critique is not a practical option.

§5.1 A Human in the Loop Is Not Calibration in the Loop

That an individual cannot instantiate an organizationally independent oversight body within their own cognitive system does not mean that the individual cannot verify facts, identify contradictions, or reject erroneous suggestions. The difference between the two is the difference between local judgment capacity and sustained institutional guarantee. That a person can detect a particular error does not mean that they possess the conditions for stably obtaining evidence, blocking erroneous execution, demanding organizational correction, and following up on consequences. The effectiveness of human oversight therefore cannot be judged by whether a person's place is retained in the flowchart; it must be assessed by what tasks that position actually undertakes, and whether the conditions for completing those tasks exist.

Bainbridge (1983), in her analysis of industrial process control and aviation automation, revealed the temporal structure of this problem. After automation takes over routine operation, operators may lose the practice and feedback needed to maintain their skills, yet still be required to take over when anomalies occur. Anomaly tasks do not disappear along with the routine tasks; they may even demand a higher degree of proficiency than normal operation. Meanwhile, timely intervention depends on an understanding of the system's current state, and that understanding must be built up gradually through continuous observation, prediction, and operation—it cannot be automatically restored at the moment a takeover prompt appears. The resulting difficulty is not merely that the supervisor's "attention is insufficiently focused"; it is that the system reduces the opportunities for forming and maintaining competence while retaining the emergency call upon that very competence.

This mechanism cannot be taken, without examination, as an established fact of skill decay among users of generative AI; what it does specify are the conditions that ought to be studied: when the system takes over retrieval, comparison, derivation, and expression, do users still obtain enough practice and feedback to maintain the knowledge required for final review? If a person comes into contact only with tidied conclusions yet is required to bear full verification responsibility for the omitted reasoning and evidence, so-called "final review" may demand that they reconstruct the entire production process on the spot. Bainbridge (1983) also discussed improvement paths through training, interface support, and human-machine collaboration; automation does not necessarily destroy oversight capacity—the key lies in whether the saved labor is accompanied by a redesign of the remaining tasks, rather than retaining the unautomated difficulties as the human's natural duty.

§5.2 Detecting Problems, Making Judgments, and Intervening

Effective oversight comprises at least three mutually irreplaceable links: detecting anomalies, judging their significance, and changing what the system does next. Visibility supports detection; expertise supports judgment; intervention authority enables judgment to have consequences. A person may possess all three at once, or be granted only part of them. Oversight failure therefore need not take the form of "nobody found the problem"; it may equally take the form of the detector being unable to obtain further evidence, the judge being unable to call for a pause, or the pause authority being written into the procedure yet punished in actual performance review.

The confirm button in a technical interface can only prove that an operation was performed; it cannot by itself prove that the operator understood all of its consequences. Nor does a veto button automatically amount to an effective right of veto. If rejecting a system suggestion requires additional explanation while accepting

it requires none; if pausing means the individual bears the loss of progress while the benefit of continuing execution accrues to team performance—then the two operations, though juxtaposed on the interface, carry different costs within the organization. At this point, whether oversight is adequate cannot be judged from confirmation rates or coverage alone; one must also examine whether objections can form and enter decision-making without being preemptively suppressed.

This also gives "independence" a more concrete meaning than the separation of personnel. That supervisor and executor are not the same person is only a starting point; if both depend on the same unverified system explanation, or if the supervisor cannot access records sufficient to challenge that explanation, the separation of personnel may still be accompanied by evidential dependence. Conversely, independent verification does not require every user to master the entire training corpus or to rebuild the whole system. For a given task, raw materials, version records, appropriate comparison conditions, and an executable pause mechanism may already suffice to support effective intervention. The required conditions should be determined by the task and its consequences—not by a false binary choice between "complete transparency" and "one can only trust."

§5.3 Control Dispersed, Responsibility Concentrated

Elish (2019) proposed the "moral crumple zone," describing a responsibility mismatch in complex automated systems: control is dispersed among designers, managers, technical systems, and operators, yet after failure, blame may concentrate on those closest to the scene of the accident whose actual control was limited. This concept is not merely another way of saying "scapegoating." It emphasizes that system boundaries, task allocation, and the mode of attributing responsibility can jointly sustain a structure in which the technology is seen as running as designed, while the human is seen as having failed to complete the remedial task that the design left to them.

In her analysis of the Three Mile Island accident, Elish (2019) does not deny that the operators' actions aggravated the accident; she points out that treating those actions as isolated events would omit the information provided by the interface, the maintenance arrangements, managerial decisions, and regulatory conditions. She is equally explicit that the moral crumple zone is not an inherent property of all complex automated systems, and she cites the attribution of responsibility to the manufacturer and software defects in the Therac-25 accident as a contrast. Her research concerns the cultural cognition and public narratives of responsibility, not a complete adjudication of legal liability in specific cases. The concept therefore supports not "operators should never be held responsible," but rather: the existence of operator fault cannot substitute for the investigation of responsibility at other nodes of control.

In AI applications, a concrete arrangement must be examined accordingly: does the organization make "the user shall verify the output" a deployment condition, yet fail to provide, in step, the materials, time, and authority that verification requires? Where such an arrangement exists, human oversight may serve two functions at once: supplementing system capacity at runtime, and providing a clearly identified object of blame after failure. The two functions are not always aligned. Even where some errors are indeed caught by humans, the allocation of responsibility may remain unbalanced, because the supervisor's having averted part of the loss does not mean that they controlled all the remaining risk. Nor can the punishment of individual users establish that system design, deployment decisions, and organizational incentives have been brought to account.

§5.4 The Reproduction of Verification Costs and Defensive Capacity

From the productivity metrics of the deploying party, the time the system takes to generate results is usually easy to record; the labor of understanding, verifying, correcting, and handling disputes may instead be dispersed between users and those affected. If evaluation counts only the generation stage, these costs are omitted. A system can shorten production time technically while transferring part of the verification labor outward organizationally. Whether this transfer occurs, and at what scale, must be confirmed by comparing complete workflows—it cannot be inferred from model output speed alone; but neither can the cost be treated as having vanished merely because it never entered the deploying party's books.

The allocation of verification costs will, in turn, alter the visible distribution of errors. Raising an effective objection usually requires keeping records, obtaining raw materials, demonstrating the discrepancy, and sustained participation in dispute processing. If these costs fall mainly on those affected, then the absence of complaints may contain both "no problem" and "unable to prove the problem." Complaint rates therefore should not be interpreted directly as reliability indicators, detached from the conditions of evidence access. When the organization then invokes a low complaint rate as proof that oversight is adequate, the unborne verification costs may participate in producing an appearance of low risk. This is a feedback mechanism requiring empirical examination, not an unconditional judgment about all AI services.

Within the MCAF framework, these conditions concretely specify how meta-audit capacity (A) is maintained or eroded within organizational relations—but authority, budgets, headcount, and skills should not be summed, undefined, into a new indicator. Training can raise recognition capacity, yet cannot automatically confer pause authority; open records can improve verification conditions, yet cannot automatically provide the time to process those records. The object of dynamic calibration is therefore not only "how much oversight capacity" exists at a given moment, but also whether the deployment process continually reproduces these capacities: whether practice is retained, whether knowledge is updated, whether objections receive responses, and whether verification labor has stable resources.

Individual verification and institutional checks are thereby not mutually exclusive proposals. Individuals can identify specific problems; organizations can pool dispersed experience, preserve evidence, and give corrections sustained force. The real asymmetry arises once the two are pulled apart: the individual retains the duty of final responsibility while losing the conditions that would make judgment effective. The construction of the defense side therefore cannot take "retaining human review" as its endpoint; it must give the person who bears responsibility capacities and conditions of control commensurate with the task, and keep the organizations that design and deploy those conditions within the range of accountability.

§5.5 The Right to Audit Local Files: The Engineered Execution of Meta-Audit at the Smallest Operable Scale

The diagnosis cited from Hodent (2017) in §5 has already revealed a structural displacement: vendor-side UX engineers the "elimination of user discomfort" into a standard procedure for development teams. This engineering has a dual problem—whether the user side can, with minimal technical means, conduct an equally systematic audit of the data extraction pipelines that the vendor embeds into the local machine. This problem does not lie within the spectrum of "cultivating digital literacy" or "reading through privacy policies"; it occurs at a lower material substrate: the local file system. Meta-audit capacity (A) at this layer is not a trained cognitive

habit but a set of reproducible operating procedures—namely, what this section terms "the right to audit local files."

Hierarchical Prefatory Declaration: The diagnosis of §5 is at the institutional level—establishing an oversight body independent of the overseen object; this organizational condition cannot be instantiated within the individual cognitive system. The action of this section, by contrast, is at the operational level—the individual need not instantiate an independent oversight body internally but need only refrain from surrendering the already-existing right to audit the file system. The two are diagnoses at different levels; the validity of the former does not negate the engineering effectiveness of the latter.

Two third-party technical audits from 2025 laid this layer's factual foundation on the table. *What Data Do Coding Agents Send, and Where To?*, the horizontal report published by Chaser Systems in November 2025, intercepted the network traffic of multiple mainstream AI coding agents under standardized tasks, revealing the structure and context of prompts—including the file paths read or modified by tool calls—and the transmission patterns and typical events of telemetry (Chaser Systems, 2025). *Unveiling Trae: ByteDance's AI IDE and Its Extensive Data Collection System*, published by Unit 221B in the same year, reverse-engineered ByteDance's AI programming environment Trae through network traffic interception, binary analysis, and runtime monitoring, identifying a data collection system comparable to enterprise-grade telemetry platforms: multiple communication channels maintain persistent connections, transmitting telemetry at a rhythm of roughly every thirty seconds even when the user is completely idle; a local WebSocket channel records the complete content of every opened file and every edit; complete file content is further packaged as "snapshots" and relayed through local microservices, forming a second channel for the wholesale transmission of code; device fingerprints based on hardware identifiers keep tracking alive even across application reinstallation (Unit 221B, 2025). More critical still is its remote feature gate system: ByteDance can enable or disable features remotely, without pushing application updates. This is precisely a ready instance of the "function creep" defined by Koops (2021)—the proper activity of a data processing system undergoes a qualitative change imperceptibly, and precisely because the change is imperceptible, it is contestable (Koops, 2021). The "imperceptibility" here precisely corresponds to a technical context: functional changes reach the user without any application update—nothing needs to be "hidden"; the change itself occurs outside the user's field of view. The continuous collection of complete file content and all editing records is one of the structural reasons why the targeting precision of the projection end Λ can keep improving, invisibly to the user, over long periods of time.

The reason this asymmetry can be long sustained at the individual scale lies in four observable mechanisms. The first is the normalization of A/B testing (online controlled experiments)—large platforms run tens of thousands of such experiments every year, with a single experiment often involving millions of users (Kohavi et al., 2020); these experiments are conducted without the users' knowledge, and the implicit consent granted by terms of service is not the same as informed consent for a specific experiment (Benbunan-Fich, 2017)—which batch of users the vendor tests what on, by what metric it converges, and when the experiment ends cannot be read by the user from any agreement text. The second is that options such as "share usage data to improve the product" are enabled by default, forming a dual payment structure under a single installation: money purchases the product; data sustains product iteration; Casilli's (2019) "click labor" (travail du clic) places local AI tools within the same spectrum as digital labor platforms—every acceptance, rejection, or modification of an AI suggestion by the user constitutes a form of unpriced labor, and the user's data feedback objectively participates in the product iteration of the model without ever being counted as labor value (Casilli, 2019). The third is

that privacy defaults are unilaterally rewritten in policy updates—Chaser Systems recorded during testing that Anthropic changed the privacy policy of Claude Code from privacy-first to opt-out, with an explicit opt-out deadline (Chaser Systems, 2025). The fourth is that application-level opt-out switches cannot fully block data exfiltration—the same report's testing of Copilot showed that telemetry requests continued to be issued after the user opted out (Chaser Systems, 2025). The superposition of the four mechanisms concretizes the institutional asymmetry diagnosed in §5, at the individual scale, into an observable time series: each unilateral change of a default is a silent reset of targeting precision.

What this section advocates is not that "the individual can solve the macro-level mapping function monopoly"—§5 has already ruled out this possibility—but rather: on the scale where the cognitive Reynolds number (Re_c) is already so high that the user cannot audit the content stream item by item, there still exists a set of minimal technical actions by which meta-audit capacity (A) is not entirely surrendered, namely, making the targeting precision of the projection end Λ partially visible.

From this, three minimal-cost operations on the user side can make the targeting precision of Λ partially visible: first, immediately after installing an AI coding or conversational tool, enter the settings panel and turn off all default switches of the form "help improve the product," "send usage data," or "allow model training"; second, after every major version update or policy change, re-check whether the above options remain turned off; third, periodically audit the tool data directories under `%APPDATA%` (Windows), `~/.config`, `~/.local/share` (Linux/macOS), observing the growth of data files, changes in configuration, and the frequency of outbound logs. These three actions do not cancel the engineering value of AI tools, nor do they promise to resolve the macro-level mapping function monopoly. They contract the macro asymmetry diagnosed in §§3–5 of this chapter to the layer of the smallest executable actions at the individual scale—i.e., the concrete evidence, at the lowest technical threshold, of the fact that "the user has not yet surrendered their own meta-audit right."

§6 Falsification Conditions

The mechanistic premise of this chapter—that indiscriminate recursive use of generated data can produce distribution degeneration under specific protocols—is supported by the controlled research of Shumailov et al. (2024). Its scope and strength in the real training ecology remain subject to examination rather than being exempted from falsification as an established fact. The three core framework diagnoses of §§3–5 can be falsified under the following conditions:

1. **Manifestation of Shadow Debt** (falsifying §3): If the AI domain experiences a concentrated settlement event similar to a financial crash (e.g., a large-scale AI accident triggering emergency regulatory intervention), and an effective compensation loop is established after that event, then this chapter's diagnosis of "no endogenous manifestation trigger" is partially falsified. Test method: observe whether an AI accident leads to the establishment of a compensation mechanism at the institutional level (e.g., an industry fund, mandatory insurance, a class-action precedent).

2. **The Ordering of the Three Clocks** (falsifying §4): If the ratio of the governance clock to the techno-capital clock (τ_g/τ_c) continuously shrinks and ultimately becomes less than 1 (i.e., the governance clock becomes faster than the techno-capital clock), then the Collingridge Dilemma has been overcome in the AI domain, and this chapter's diagnosis of "persistent concealment" does not hold. Test method: track the change in the ratio of the model release cycles of major AI labs to the passage cycles of AI legislation

in major jurisdictions.

3. **The Absence of Counterbalance at the Individual Level** (falsifying §5): If it can be demonstrated that individuals, through some form of cognitive training or technical assistive tool, can continuously maintain effective meta-audit capacity in daily use (i.e., the rate of decline of individual effective meta-audit capacity is lower than the growth rate of Λ), then the diagnosis of "institutional asymmetry at the individual level" in §5 of this chapter is falsified. Test method: in a controlled experiment, have one group of users undergo cognitive training and use technical assistive tools, measure the rate of decline of their meta-audit capacity over time, and compare it with the contemporaneous growth rate of Λ (using changes in response latency for the same product interaction cycle or the decreasing trend in the frequency of decision prompt insertions as proxy indicators).

   An Honest Annotation of Operational Difficulties. The above falsification conditions face two difficulties at the operational level: (1) the acceleration of the governance clock may be offset by the expansion of the scope of governance—the speed of legislation increases, but the coverage of regulation is also expanding; the direction of the change in the ratio is unclear; (2) the establishment of a compensation mechanism requires identifying "AI accidents," and "AI accidents" themselves lack a legal definition. Therefore, these falsification conditions are currently closer to a "contestable framework" than to "testable hypotheses." Honest annotation of this state is itself a requirement of the framework's self-audit.

§7 Conclusion: What We Have Come to Know

This chapter has performed a dynamic calibration of the AI domain using the MCAF framework. The arrangement of the core findings is not "what we did" but "what we have come to know":

On the empirical premise, the model-collapse research of Shumailov et al. (2024, *Nature*) establishes the risk of degeneration under specific recursive-training protocols: when model outputs indiscriminately replace original data and become inputs for subsequent training, information in the distribution tail can be irreversibly lost. This chapter uses that result to examine how the return of public corpora increases pressure on unlocked stock (C), but it does not treat the claim that "AI training-data contamination never contracts" as a universal fact established by that study. The effects of retaining and accumulating real data and of external verification must be assessed along each specific path. On this empirical premise, this chapter presents three intra-framework diagnoses:

First, the concealment phase lacks the constraint of an institutionalized manifestation mechanism (§3). AI shadow debt has no financial-style endogenous manifestation trigger—no automatic chain of underlying-asset default, margin call, and rating downgrade. At present, there is neither automatic market settlement nor a stable mechanism of compulsory policy settlement.

Second, while current institutional and technological conditions persist, the defense side faces long-term lag pressure (§4). The techno-capital period is shorter than the governance period, while the debt-manifestation period is highly uncertain and generally longer; this is not a one-off temporal gap but a tendency continually reproduced by the present structure. More critically, AI governance has not developed the stable emergency-compression capacity found in the financial system's crisis mode.

Third, the individual cannot resist algorithmic sycophancy (§5)—not because of insufficient willpower, but be-

cause the institutional conditions for counterbalance (establishing an oversight body independent of the overseen object) do not hold at the individual level.

These three diagnoses (together with the empirical premise on which they rest) converge on a single conclusion: AI's shadow debt may be harder to track and harder to settle than financial shadow debt. This is not to say that AI is "more dangerous" than finance—it is that the debt dynamics of AI are structurally different from those of finance, and we have yet to develop auditing tools that match this difference. The contribution of this chapter is not to "solve" this structural predicament but to honestly annotate it. Making the invisible accumulation of debt visible is the most fundamental work the MCAF framework can do in the AI domain. Taken together, the three diagnoses are precisely the chapter title written in dynamics—the projection end expands on a monthly scale, the defense end arrives years late, and the settlement has no timetable; these are the current readings of the three clocks in the clock room.

References

- Bainbridge, L. (1983). Ironies of Automation. *Automatica*, 19(6). https://doi.org/10.1016/0005-1098(83)90046-8
- Bell, M. L. & Davis, D. L. (2001). Reassessment of the lethal London fog of 1952: Novel indicators of acute and chronic consequences of acute exposure to air pollution. *Environmental Health Perspectives*, 109(suppl 3). https://doi.org/10.1289/ehp.01109s3389
- Bell, M. L., Davis, D. L., & Fletcher, T. (2004). A retrospective assessment of mortality from the London smog episode of 1952: The role of influenza and pollution. *Environmental Health Perspectives*, 112(1). https://doi.org/10.1289/ehp.6539
- Benbunan-Fich, R. (2017). The ethics of online research with unsuspecting users: From A/B testing to C/D experimentation. *Research Ethics*, 13(3-4). https://doi.org/10.1177/1747016116680664
- Brimblecombe, P. (1987). *The Big Smoke: A History of Air Pollution in London since Medieval Times*. London: Methuen.
- Casilli, A. A. (2019). *En attendant les robots: Enquête sur le travail du clic*. Paris: Seuil.
- Chaser Systems. (2025). *What Data Do Coding Agents Send, and Where To?* https://chasersystems.com/blog/what-data-do-coding-agents-send-and-where-to/
- Corton, C. L. (2015). *London Fog: The Biography*. Cambridge, MA: The Belknap Press of Harvard University Press.
- Cyberspace Administration of China et al. (2023). *Interim Measures for the Management of Generative Artificial Intelligence Services*. Chinese Government Website. (In Chinese.) https://www.gov.cn/zhengce/zhengceku/202307/content_6891752.htm
- Davis, D. (2002). *When Smoke Ran Like Water: Tales of Environmental Deception and the Battle Against Pollution*. New York: Basic Books.
- Elish, M. C. (2019). Moral Crumple Zones: Cautionary Tales in Human-Robot Interaction. *Engaging Science, Technology, and Society*, 5. https://doi.org/10.17351/ests2019.260
- Friestad, M. & Wright, P. (1994). The Persuasion Knowledge Model: How People Cope with Persuasion Attempts. *Journal of Consumer Research*, 21(1). https://doi.org/10.1086/209380
- Greater London Authority (GLA). (2002). *50 Years On: The Struggle for Air Quality in London since the Great Smog of December 1952*. London: GLA. https://cleanair.london/app/uploads/CAL-217-Great-

Smog-by-GLA-20021.pdf
- Hodent, C. (2017). *The Gamer's Brain: How Neuroscience and UX Can Impact Video Game Design*. Boca Raton, FL: CRC Press (Taylor & Francis Group).
- Kohavi, R., Tang, D. & Xu, Y. (2020). *Trustworthy Online Controlled Experiments: A Practical Guide to A/B Testing*. Cambridge: Cambridge University Press.
- Koops, B.-J. (2021). The concept of function creep. *Law, Innovation and Technology*, 13(1). https://doi.org/10.1080/17579961.2021.1898299
- Marjanovic, O., Cecez-Kecmanovic, D., & Vidgen, R. (2021). Algorithmic pollution: Making the invisible visible. *Journal of Information Technology*, 36(4). https://doi.org/10.1177/02683962211010356
- Moody's Analytics. (2025). *Quarterly Commercial Real Estate Briefing: Recovery on Pause* (August 2025). https://events.moodys.com/mc71247-wbn-2025-mau27645-q2-cre-qeb
- Muldoon, J. & Parke, J. J. (2025). Cruel companionship: How AI companions exploit loneliness and commodify intimacy. *New Media & Society*. https://doi.org/10.1177/14614448251395192
- Shumailov, I. et al. (2024). AI models collapse when trained on recursively generated data. *Nature*, 631. https://doi.org/10.1038/s41586-024-07566-y
- Stanford HAI. (2026). *The 2026 AI Index Report*. Stanford University. https://hai.stanford.edu/ai-index/2026-ai-index-report
- Unit 221B. (2025). *Unveiling Trae: ByteDance's AI IDE and Its Extensive Data Collection System*. https://blog.unit221b.com/dont-read-this-blog/unveiling-trae-bytedances-ai-ide-and-its-extensive-data-collection-system
- Weizenbaum, J. (1976). *Computer Power and Human Reason: From Judgment to Calculation*. San Francisco: W. H. Freeman.
- Zhang, Sheng. (2017). "The Evolution of the Ancient Chinese Oversight System: From a Composite System to a Unitary System." *Administrative Law Review*, 2017(4). (In Chinese.) http://iolaw.cssn.cn/fxyjdt/201707/t20170730_4651843.shtml
- Zheng, J. & Meister, M. (2025). The unbearable slowness of being: Why do we live at 10 bits/s? *Neuron*, 113(2). https://doi.org/10.1016/j.neuron.2024.11.008